

Predicting Backup Non-Compliance Using KPI-Derived Labels and Machine Learning: A Proof-of-Concept Study

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Abstract. Ensuring backup compliance is essential for data protection and operational resilience in enterprise IT environments. This study proposes a KPI-aware machine learning approach combining weak supervision, rule-based labeling, and supervised learning to estimate KPI-derived backup non-compliance status. The approach was evaluated on 439 device instances with 33 features using an 80:20 stratified split, producing a test set of 88 instances comprising 86 non-compliant and only 2 compliant proxy labels, indicating extreme class imbalance. Proxy labels were generated from predefined KPI rules without manual annotation or independent compliance audits. Logistic Regression, Random Forest, and XGBoost were evaluated using AUC-PR, F1-score, recall, Brier score, and calibration analysis. Random Forest achieved the strongest comparative results, with an AUC-PR of 0.998, F1-score of 0.994, recall of 1.000, and the lowest Brier score of 0.014. However, calibration remains uncertain because only two compliant instances were available. Therefore, these results indicate preliminary feasibility for reproducing or estimating KPI-derived proxy labels and should not be interpreted as independent validation of real-world non-compliance detection. The approach provides a scalable basis for backup compliance monitoring and preliminary risk prioritization in enterprise IT systems.

Keywords: Backup Non-Compliance, KPI-Derived Labels, Machine Learning, Label Leakage, Enterprise IT

1. INTRODUCTION

In modern enterprise IT environments, ensuring reliable data protection through backup systems [1] is a fundamental requirement for maintaining operational resilience [2] and business continuity. Failures in backup processes can lead to critical loss of data, service disruption, and non-compliance with organizational policies [3], [4]. However, as enterprise systems grow in scale and complexity, maintaining consistent backup compliance across heterogeneous infrastructures become increasingly challenging [5]. Traditional monitoring approaches rely heavily on rule-based mechanisms and periodic audits. These methods are often static, reactive, and unable to cope with the volume, velocity, and heterogeneity of modern enterprise data, leading to delayed detection of compliance violations and operational failures [3]. Such limitations create a significant gap between risk occurrence and detection capability, preventing organizations from responding proactively to emerging issues [6].

The rapid evolution of cloud-based infrastructures further exacerbates these challenges. Conventional governance models, which depend on manual processes and predefined thresholds, lack adaptability and fail to provide real-time insights into system behavior [7]. As a result, organizations face increased exposure to operational risks, security threats, and compliance violations [8]. To address these limitations, Artificial Intelligence (AI) and Machine Learning (ML) have emerged as key enablers of intelligent monitoring systems. ML techniques allow automated analysis of large-scale data, enabling anomaly detection, classification, and predictive modeling of system behavior. These capabilities significantly improve early warning systems by identifying hidden patterns and predicting potential failures before they occur [6], [8]. In particular, Key Performance Indicators (KPIs) have become essential components for monitoring and evaluating system performance [3]. KPIs provide quantitative representations of system states and are widely used across domains such as manufacturing and telecommunications for anomaly detection and performance optimization [9], [10] a pattern this study extends to backup compliance monitoring in enterprise IT systems. However, KPI data is inherently complex, often represented as time-series with dynamic patterns, contextual dependencies, and non-linear relationships, making anomaly detection a non-trivial task [11], [12].

Traditional threshold-based monitoring systems are insufficient for capturing such complexity, as anomalies may manifest in multiple forms, including point anomalies,

collective anomalies, and contextual anomalies [11]. Furthermore, statistical approaches often fail to model the non-linear interactions present in real-world systems, especially when dealing with high-dimensional and heterogeneous data [10]. Recent studies demonstrate that machine learning models, particularly ensemble methods such as Random Forest and Gradient Boosting, are highly effective in handling complex KPI-driven systems. These models can learn intricate patterns from data and outperform traditional statistical approaches in predictive tasks across various domains such as predictive maintenance and enterprise risk modeling, both of which share the class-imbalance challenge central to backup compliance monitoring [10], [13]. Moreover, modern monitoring paradigms emphasize the importance of real-time and data-driven approaches. Online KPI monitoring enables continuous analysis of system behavior and supports proactive decision-making. Within the context of IT operations monitoring, backup status, agent availability, and backup recency constitute important operational indicators for identifying devices that require further attention. However, implementing such systems introduces challenges such as feature engineering, handling heterogeneous data sources, data quality issues, class imbalance, and concept drift [9], [11], [13]. Another critical challenge in enterprise environments is the scarcity of labeled data. Manual annotation of anomalies or compliance violations is often costly and impractical at scale. Therefore, there is a growing need for approaches that leverage domain knowledge to generate labels automatically while maintaining high predictive performance [11].

In addition, traditional monitoring systems often fail to align technical metrics with business impact. Modern approaches in Site Reliability Engineering (SRE) advocate integrating KPI-based monitoring with machine learning to bridge this gap and enable more context-aware and business-driven decision-making [14], [15]. To address these challenges, this study proposes a KPI-aware machine learning approach for predicting backup non-compliance in enterprise IT systems. The workflow integrates domain-driven rules derived from operational KPIs with supervised machine learning techniques, enabling automatic label generation without requiring manual annotation. To guide the investigation, this study addresses three research questions: (RQ1) can machine learning models reproduce KPI derived backup compliance labels; (RQ2) which model provides the best discrimination and calibration under extreme class imbalance; and (RQ3) does the model provide operational value beyond the original KPI rule, defined here as improved risk ranking and calibrated prioritization (via AUC-PR, cumulative gain, and decision curve

analysis) rather than measured workload reduction, evaluated as preliminary evidence detailed in the Results and Discussion. It is important to clarify why machine learning remains necessary even though the target label itself is generated from a deterministic KPI rule. The KPI rule applied in this study only defines a post hoc condition for non-compliance and requires complete, real time operational data to be executed directly. Machine learning is employed to generalize patterns across combinations of KPI derived features, such as storage utilization ratio, check in recency, and primary user presence, that may correlate with non-compliance risk beyond a single threshold rule, and to produce probabilistic risk scores that support prioritization rather than a binary rule outcome alone; the rule itself, moreover, cannot rank devices by risk and requires complete real-time data to run, whereas a trained model can estimate risk from partial data and generalize to new devices. To avoid conceptual ambiguity, this study distinguishes three related but distinct notions. Rule based detection refers to the conventional approach that relies on static thresholds evaluated manually or periodically. Rule based labeling, the technique adopted in this study, applies domain rules to automatically generate target labels in place of manual annotation, rather than performing real time detection. Machine learning prediction refers to a supervised model trained on the rule derived labels and associated features, with the aim of generalizing to patterns that may extend beyond a single deterministic rule.

The proposed workflow consists of data preprocessing, feature engineering, KPI-based labeling, and machine learning model training. Multiple models, including Logistic Regression, Random Forest, and XGBoost, are evaluated using metrics suitable for imbalanced classification problems, such as precision, recall, F1-score, and AUC-PR. The main contributions of this study have three aspects. First, a KPI-driven labeling strategy for automatic detection of backup non-compliance. Second, a robust preprocessing and feature engineering pipeline for heterogeneous enterprise IT datasets. Third, a comparative evaluation of machine learning models for predicting backup compliance. It should be noted that, based on the current experimental design, the label and the input features are derived from the same time snapshot, without any temporal shift. Consequently, the model in its present form reproduces the current backup non-compliance status rather than predicting future non-compliance. This distinction is made explicit to ensure accurate interpretation of the study's contribution, and the extension toward genuine temporal prediction is discussed as a direction for future work. This

study demonstrates that integrating KPI-based domain knowledge with machine learning provides a preliminary workflow for investigating KPI-based backup compliance monitoring to improve monitoring systems and ensuring compliance in enterprise IT environments.

2. RELATED WORK

Recent studies indicate an increasing demand for intelligent monitoring systems as enterprise environments become more complex and data-intensive [16]. Traditional rule-based approaches are no longer sufficient to handle the dynamic and heterogeneous nature of modern data. Parimi et al. demonstrated that machine learning models such as Logistic Regression, Random Forest, and Gradient Boosting can effectively predict compliance violations and operational risks, enabling a transition from reactive to proactive monitoring [6]. Similarly, S. Thota. proposed an AI-based data governance approach for cloud systems, emphasizing real-time anomaly detection, automated compliance validation, and scalability [8]. In KPI-based monitoring, Gram et al. utilized real-time KPI streams for anomaly detection, while Cil nio et al. introduced an adaptive workflow that clusters KPIs and detects different types of anomalies, including point, collective, and contextual anomalies [9], [10]. Overall, these studies highlight the limitations of static threshold-based methods and demonstrate the advantages of machine learning in handling complex and dynamic data.

Furthermore, Yuliana et al. showed that ensemble models such as Random Forest and XGBoost outperform traditional statistical methods in modeling KPI-driven system behavior, particularly in non-linear and complex environments [11]. Kumar et al. reinforced this finding by proposing a hybrid machine learning approach that combines domain knowledge with data-driven techniques to improve prediction accuracy and robustness in anomaly detection tasks [13]. Meanwhile, Kolluri et al. emphasized the importance of aligning technical metrics with business impact through KPI-based anomaly detection within the context of Site Reliability Engineering (SRE) [14]. Despite these advancements, most existing studies focus on domains such as telecommunications, manufacturing, and energy systems, with limited attention to backup compliance in enterprise IT environments. Additionally, many approaches rely on unsupervised methods without incorporating domain-driven labeling strategies. Existing studies further illustrate this

gap. Parimi and Yallavula [6] proposed a unified enterprise risk intelligence approach covering compliance, fraud, and operational risk, but relied on already labeled fraud cases and unsupervised methods for rare compliance events, and reported that misclassifications were linked to data quality issues and concept drift. Thota [8] focused on cloud data governance for Salesforce-based enterprises using automated classification and unsupervised models, without an explicit labeling strategy or reported limitations. Kumar et al. [13] applied a rule-based threshold to generate health-status labels for predictive maintenance in lithium-ion batteries, closely resembling the KPI-based labeling approach adopted in this study, though in a different domain, and noted that model performance remained highly dependent on input data quality. Kolluri [14] introduced contextual anomaly detection for business KPIs within a Site Reliability Engineering context, combining supervised and unsupervised approaches but without reporting explicit study limitations. As shown, prior studies primarily target enterprise risk, cloud governance, predictive maintenance, or business KPI domains, and largely rely on directly available, unsupervised, or hybrid labels (as summarized in Table 1). In contrast, the present study focuses specifically on backup compliance in enterprise IT systems and adopts a rule based (weak supervision) labeling strategy to construct supervisory signals in the absence of independently audited ground truth. This positioning clarifies that the primary novelty of this work lies in the domain application and labeling strategy, rather than in the underlying machine learning algorithms, which are established methods widely used across other domains.

Table 1. Comparison of Prior Studies on KPI-Based Monitoring and Compliance Prediction

Study	Domain	Dataset	Labeling Strategy	Model(s)	Reported Limitation
Parimi & Yallavula [6]	Enterprise risk (compliance, fraud, operational risk)	Not explicitly reported	Pre-labeled fraud cases combined with unsupervised methods for rare compliance events	Logistic Regression, Random Forest, Gradient Boosting	Misclassification linked to data quality issues and concept drift
Thota [8]	Cloud data governance (Salesforce)	Not explicitly reported	Automated classification and unsupervised methods; no explicit labeling strategy	Not specified	Not reported
Gram et al. [9]	Manufacturing (real-time KPI streams)	Not explicitly reported	Not specified	Not specified	Not reported
Cil�nio et al. [10]	Mobile networks (KPI clustering)	Not explicitly reported	Not specified	Clustering-based	Not reported

Yuliana et al. [11]	Telecommunications (base station traffic)	Not explicitly reported	Not specified	Random Forest, XGBoost	Not reported
Kumar et al. [13]	Predictive maintenance (lithium-ion batteries)	Not explicitly reported	Rule-based threshold for health-status labels	Enhanced Random Forest	Performance highly dependent on input data quality
Kolluri [14]	Business KPI (Site Reliability Engineering)	Not explicitly reported	Combined supervised and unsupervised approaches; no explicit strategy	Not specified	Not reported
This study	Backup compliance, enterprise IT systems	439 devices, 33 features	Rule-based (weak supervision)	Logistic Regression, Random Forest, XGBoost	Extreme class imbalance, single-organization dataset, potential label leakage, no temporal validation

Therefore, this study addresses these gaps by proposing a KPI-aware machine learning workflow that integrates domain knowledge for automatic label generation and applies supervised learning to predict backup non-compliance in enterprise IT systems.

3. METHODS

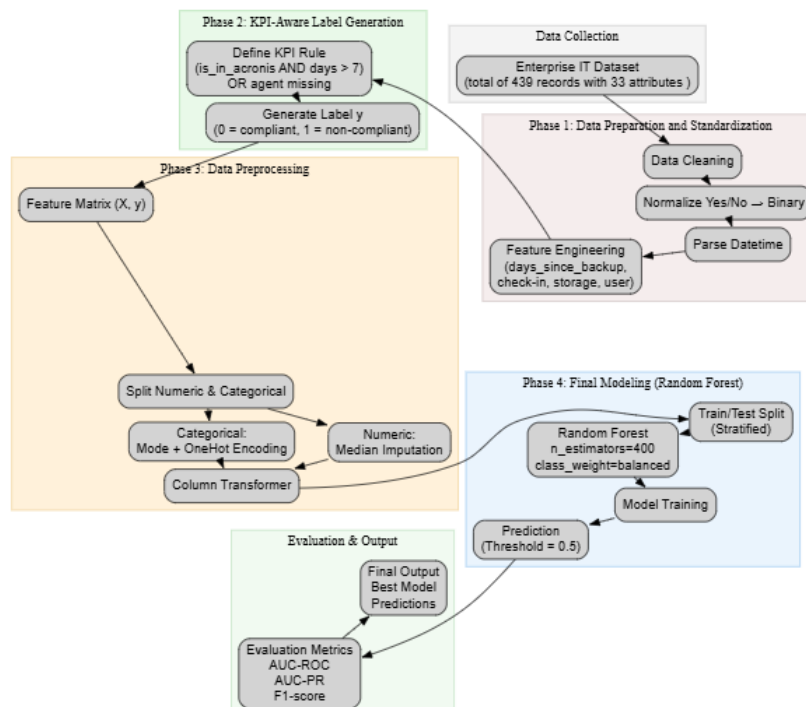


Figure 1. Architecture System

Figure 1 illustrates the proposed methodological workflow, consisting of six stages: data collection, data preparation and standardization, KPI-aware label generation, data preprocessing, modeling, and evaluation. The study uses an enterprise IT dataset comprising 439 device instances and 33 attributes. Data preparation includes cleaning, categorical normalization, datetime parsing, and feature engineering to derive operational indicators such as backup and system check-in recency.

The KPI-aware labeling stage generates the target variable from predefined operational rules, labeling devices as non-compliant when the backup threshold is exceeded or the backup agent is unavailable. The data is then preprocessed through feature construction, missing-value imputation, and one-hot encoding. Logistic Regression, Random Forest, and XGBoost are trained using a stratified train-test split and evaluated using complementary classification metrics. The resulting model outputs are used to assess backup non-compliance.

3.1. Data Collection

This study utilizes an enterprise IT dataset comprising 439 instances and 33 features, covering device information, backup records, system configuration, and operational status [17], [18]. The dataset includes attributes from Entra, Intune, Kaseya, and Acronis, as well as device specifications, user information, backup status, last-backup timestamps, system check-in records, storage capacity, and agent configuration [18], [19]. The data are heterogeneous, consisting of numerical, categorical, and datetime features, with missing values and non-standard entries such as "offline" and "unavailable". These characteristics require subsequent preprocessing and standardization before modeling [17].

3.2. Data Preparation and Standardization

This stage transforms the raw dataset into a structured format suitable for machine learning. Column names are standardized into a unified schema to ensure consistency across data sources [20], [21]. Binary categorical values such as "Yes" and "No" are converted to 1 and 0, respectively. Datetime fields are parsed to derive temporal indicators, including days since the last backup and system check-in [11]. Further transformations are applied to extract structured numerical information from semi-structured fields. Storage-related attributes, such as disk size and backup zone capacity, are converted into numerical values in gigabytes, and additional ratios are computed to

reflect storage utilization. Moreover, user-related indicators are derived by identifying the presence of a primary user, while software-related attributes, such as backup agent versions, are transformed into comparable numerical representations. These steps ensure that heterogeneous raw data is converted into meaningful and machine-readable features [6], [20], [21].

3.3. KPI-Aware Labeling

Since explicit labels were not available in the dataset, a KPI-aware labeling strategy was employed to construct the target variable automatically. The resulting labels should be interpreted as rule-derived proxy labels rather than independent ground-truth labels obtained through human annotation or operational compliance audits. In this study, backup non-compliance was defined using predefined operational KPI rules. A device was labeled as non-compliant only if it was registered in the backup system and satisfied at least one of the following conditions: the number of days since its last successful backup exceeded seven days, or its backup agent version was missing, empty, marked as offline, or marked as unavailable. Devices that did not satisfy these conditions were labeled as compliant. The seven-day threshold was predefined in the labeling procedure and was not estimated or learned from the dataset [22]. This labeling procedure can be viewed as a form of weak supervision (also referred to in the literature as programmatic labeling), in which predefined domain rules are used to generate proxy labels in the absence of independently annotated ground truth [23], [24]. The labeling rule is formally defined as shown in Equation (1).

$$y_i = \begin{cases} 1, & \text{if } (A_i = 1 \wedge D_i > \tau) \vee (M_i = 1) \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

Where y_i denotes the compliance label of device i , A_i is a binary indicator representing whether the device is registered in the backup system, D_i represents the number of days since the last successful backup, τ denotes the predefined threshold (set to 7 days in this study), and M_i is a binary indicator representing the absence or unavailability of the backup agent.

3.4. Data Preprocessing

This stage prepares the dataset for model training by transforming features into a structured format suitable for machine learning algorithms. The preprocessing pipeline

begins with feature selection, where relevant attributes derived from the previous stages are used to construct the feature matrix X and target vector y . The complete feature matrix X comprises 20 attributes, grouped into four categories: (i) platform-integration indicators (`is_in_entra`, `is_in_intune`, `is_in_kaseya`, `is_in_acronis`); (ii) temporal/freshness indicators (`days_since_kaseya_checkin`, `days_since_last_backup`, `has_primary_user`); (iii) Acronis configuration and storage indicators (`acronis_ocr`, `acronis_asm`, `acronis_uefi_correct`, `acronis_cert`, `disk_size_gb`, `asz_size_gb`, `asz_free_gb`, `asz_free_ratio`); and (iv) categorical device/context attributes (`operating_system`, `device_model`, `country_code`, `location`, `kaseya_region`). The selected attributes are used to construct the feature matrix X and target vector y . The feature matrix retains the operational variables available in the dataset, including `is_in_acronis` and `days_since_last_backup`, which are also involved in the KPI-based labeling procedure. Therefore, the predictive models may learn patterns that are directly or indirectly related to the rule-derived target. This dependency is considered when interpreting the model performance, particularly because the target label is generated from the same operational snapshot used to construct the predictor variables. No separate leakage-controlled feature variant is implemented in the experimental pipeline. It is also important to note that the label in Eq. (1) reflects the device's non-compliance status at the snapshot time used to compute the backup-recency variable, that is, a current, cross-sectional target, and the workflow does not predict future non-compliance. Extending it to a genuinely predictive, forward-looking target would require temporally shifting the label relative to the feature snapshot, for example by using features observed up to time t to predict non-compliance at $t+k$. This is left for future work, in line with recommendations for hybrid domain-knowledge and data-driven predictive workflow [13]. Given the heterogeneous nature of the dataset, features are divided into numerical and categorical groups. Numerical features are handled using median imputation to address missing values, which provides robustness against outliers. Formally, the imputation process is defined as:

$$x_{ij} = \begin{cases} x_{ij}, & \text{if observed} \\ \tilde{x}_j, & \text{if missing} \end{cases} \quad (2)$$

In Equation (2), x_{ij} denotes the value of the $-j$ th feature for the $-i$ th instance x_{ij} , denotes the resulting value after imputation, and $\tilde{x}_j$ represents the median of feature calculated from the observed values. Categorical features are first converted into a consistent string format to ensure uniform representation. Missing categorical values are then

imputed using the most frequent category (mode), which preserves the dominant pattern in the data. After imputation, one-hot encoding is applied to transform categorical variables into binary indicator vectors. Each category is represented as a separate binary feature, where the value indicates the presence or absence of that category. This transformation can be expressed as:

$$z_{ik} = \begin{cases} 1, & \text{if } x_i \in C_k \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

In Equation (3), z_{ik} denotes the binary indicator for category k of instance i , represents the original categorical value of instance i , and C_k denotes the k -th category. The value of z_{ik} is 1 when x_i belongs to C_k , and 0 otherwise [24]. All preprocessing steps are integrated using a column-wise transformation pipeline, where numerical and categorical features are processed according to their respective characteristics. This unified pipeline ensures that the final dataset is fully numerical, consistent, and free from missing values, thereby enabling effective and reliable model training while preserving the informational structure of the original data [26], [27]. After preprocessing, the resulting feature matrix contains 63 predictors derived from 20 selected attributes. The training and testing sets have dimensions of 351×63 and 88×63 , respectively. These 63 predictors are subsequently used as inputs for the machine learning models.

3.5. Model Development and Evaluation

A ColumnTransformer pipeline was constructed to process the heterogeneous feature space, where numerical attributes were handled using median imputation, while categorical features were imputed using the most frequent value and subsequently transformed via one-hot encoding [10], [25].

The dataset was partitioned using stratified sampling with an 80:20 ratio, resulting in 351 training instances and 88 testing instances. The full dataset consisted of 430 non-compliant (97.95%) and 9 compliant (2.05%) devices, while the test set contained 86 non-compliant and 2 compliant devices. The resulting class distribution is presented in Table 2.

Table 2. Class Distribution of label_backup_noncompliance

Split	N	Non-compliant (label = 1)	Compliant (label = 0)
Full dataset	439	430 (97.95%)	9 (2.05%)

Training set (80%)	351	344 (98.0%)	7 (2.0%)
Test set (20%)	88	86 (97.7%)	2 (2.3%)

Because of the extreme class imbalance and the limited number of compliant instances, a single hold-out split may provide an unstable performance estimate. Therefore, model comparison was additionally performed using repeated stratified 5-fold cross-validation with 10 repetitions. For each evaluation metric, the mean and 95% confidence interval were calculated from the 50 validation scores obtained across the 5 folds and 10 repetitions. The confidence interval was estimated using the normal approximation as $\left(\bar{x} \pm 1.96(s/\sqrt{n})\right)$, where $(\bar{x})$ is the mean score, (s) is the standard deviation, and $(n=50)$. The experiments were conducted using Python 3.13.15 with scikit-learn 1.6.1, XGBoost 3.4.1, pandas 2.2.3, and NumPy 2.1.3. A fixed random seed of 42 was applied to support reproducibility.

Three classifiers were evaluated within a unified pipeline to ensure a fair comparison. A majority-class baseline was incorporated to establish a reference level of performance under the highly imbalanced class distribution. Because this baseline may yield deceptively high accuracy while failing to identify the rare compliant class, model evaluation additionally considered balanced accuracy, macro F1-score, and class-specific performance measures [4].

Three predictive machine learning models were evaluated within the unified pipeline: Logistic Regression was employed as a linear baseline with `max_iter=2000` and `class_weight="balanced"` [28]. Random Forest used `n_estimators=400` and `class_weight="balanced_subsample"`, while XGBoost used 600 estimators, a learning rate of 0.05, a maximum depth of 6, and L2 regularization (`reg_lambda=1.0`). The hyperparameters were manually selected based on preliminary experiments and configurations commonly used in related studies. No automated hyperparameter optimization was applied [29]. All remaining hyperparameters not explicitly specified above were retained at their default scikit-learn/XGBoost values. Class weighting was adopted across all models to address class imbalance without introducing synthetic data.

In addition to the majority-class baseline, the Original KPI Rule was retained as a rule-based sanity-check baseline. It applies the deterministic KPI rule described in Equation

(1), is not trained as a predictive model, and serves as a reference for assessing the consistency of the predictive models with the rule-derived target labels.

The positive class (label 1) represents non-compliant devices, which constitute the majority class in this dataset (430 of 439 devices; 97.95%). Therefore, AUC-PR primarily evaluates the ranking performance for the majority non-compliant class rather than the rare compliant class. Although AUC-PR remains useful for assessing discrimination of the positive class, its high values should not be interpreted as evidence of strong minority-class detection. Class-specific precision, recall, and F1-score are therefore also reported to explicitly assess performance on compliant devices.

Precision measures the reliability of positive predictions by indicating the proportion of devices predicted as non-compliant that were also labeled as non-compliant by the KPI rules. It is defined as:

$$Precision = \frac{TP}{TP+FP} \quad (4)$$

Where TP denotes correctly identified non-compliant devices, and FP represents compliant devices incorrectly classified as non-compliant. A high precision value implies a low false alarm rate. Recall measures the model's ability to retrieve devices assigned a non-compliant proxy label by the KPI-based labeling mechanism and is defined as:

$$Recall = \frac{TP}{TP+FN} \quad (5)$$

Where FN corresponds to non-compliant devices that were not detected. High recall is crucial in this context, as undetected violations may lead to operational risks. To balance the trade-off between precision and recall, the F1-score is used as their harmonic mean:

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{TP+FN} \quad (6)$$

This metric provides a single measure that reflects both correctness and completeness of the model's predictions. Beyond threshold-dependent metrics, the overall discriminative capability of the model is evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC-ROC):

$$AUC - ROC = \int_0^1 TPR d(FPR) \quad (7)$$

This metric captures the trade-off between the true positive rate (TPR) and false positive rate (FPR) across all classification thresholds [30]. However, in imbalanced classification settings, AUC-ROC may yield overly optimistic results. Therefore, this study emphasizes the Area Under the Precision–Recall Curve (AUC-PR) [31]:

$$AUC - PR = \int_0^1 Precision, d(Recall) \quad (8)$$

AUC-PR is used as an important discrimination metric for the positive class, which in this study corresponds to non-compliant devices. Because non-compliant devices constitute the majority class, high AUC-PR values should be interpreted cautiously and should not be taken as evidence of effective detection of the rare compliant class. Therefore, model selection considers AUC-PR together with class-specific precision, recall, and F1-score to provide a more balanced assessment of both classes. The model achieving the strongest overall comparative performance across these evaluation measures is selected as the final predictive model, while accuracy is not emphasized due to its limitations under skewed class distributions [30]. The proposed KPI-aware machine learning approach is systematically represented in Algorithm 1.

Algorithm 1 KPI-Aware Machine Learning Approach for Backup non-compliance Prediction

Require: Raw enterprise IT dataset (D_{raw}), backup threshold $\tau = 7$ days, Prediction threshold $\theta = 0.5$

Ensure: KPI-derived predicted labels $\hat{y}$, Risk probabilities p , Priority ranking R , KPI recommendations (K_{rec})

Phase 1: Data Preparation and Feature Engineering

Input $\leftarrow$ raw dataset D

Clean and standardize attributes, binary values, datetime, and storage fields

Derive backup recency, check-in recency, storage ratio, user, and platform indicators

Output $\leftarrow$ engineered dataset (D_{eng})

Phase 2: KPI-Aware Label Generation

Input $\leftarrow D_{eng}, \tau$

For each device d_i do

 if $is_in_acronis = 1$ AND

$(days_since_last_backup > \tau$ OR

$agent_status \in \{missing, offline, unavailable\})$ then

$y_i \leftarrow 1$ // non-compliant

 else

$y_i \leftarrow 0$ // compliant

 end if

end for

Output $\leftarrow$ KPI-derived proxy labels y

Phase 3: Data Preprocessing

Input $\leftarrow$ D_{eng}, y

Split numerical and categorical features

Apply median/mode imputation and one-hot encoding

Output $\leftarrow$ preprocessed feature matrix X

Phase 4: Model Training and Validation

Input $\leftarrow$ X, y

Perform stratified 80:20 train-test split

Train Logistic Regression, Random Forest, and XGBoost

Evaluate using repeated stratified 5-fold cross-validation

Compute AUC-PR, F1-score, recall, AUC-ROC, and Brier score

Select model M^* with the strongest comparative performance

Output $\leftarrow$ selected model M^* and evaluation metrics E

Phase 5: Risk Scoring and Prioritization

Input $\leftarrow$ M^*, X

$p \leftarrow M^*.predict_proba(X)$

$\hat{y} \leftarrow 1$ if $p \geq \theta$, otherwise 0

Sort devices by p in descending order

Select highest-risk devices as remediation priority

Output $\leftarrow$ risk probabilities p , predicted labels $\hat{y}$, priority ranking R

Phase 6: KPI Recommendation

Input $\leftarrow$ priority ranking R and device KPI values

for each high-risk device d_i do

 if $days_since_last_backup > 7$ then

 recommend "Perform or verify backup immediately"

 end if

 if Acronis agent is missing/offline/unavailable then

 recommend "Restore or reinstall the backup agent"

 end if

 if $days_since_kaseya_checkin > 30$ then

 recommend "Verify device connectivity and management-agent status"

 end if

 if storage availability is below the operational threshold then

 recommend "Review and increase available backup storage"

 end if

end for

Output $\leftarrow$ prioritized KPI recommendation list (K_{rec})

return $M^*, E, p, \hat{y}, R, (K_{rec})$

4. RESULTS AND DISCUSSION

4.1. Results

This section presents the experimental results of the proposed KPI-aware machine learning approach for predicting backup non-compliance. The test set consists of 88 instances, including 86 non-compliant and 2 compliant devices, indicating a highly imbalanced class distribution. Table 3 summarizes the core classification performance of the predictive models and reference baselines. On the single hold-out test set, Random Forest achieves the highest AUC-ROC (0.950581), AUC-PR (0.998), and F1-score (0.994220) among the predictive models, together with a recall of 1.000000. It also obtains the lowest Brier score (0.013568); however, this value should be interpreted cautiously because the test set contains only two compliant instances. Logistic Regression shows competitive but lower discriminative performance, while XGBoost exhibits the weakest class separation. As shown in Figure 2, Random Forest demonstrates the strongest discriminative performance on the single hold-out test set. However, because non-compliant devices constitute the majority positive class, the corresponding AUC-PR should be interpreted as discrimination of the majority class rather than evidence of effective detection of the rare compliant class. Given the imbalanced nature of the dataset, ROC analysis alone is insufficient.

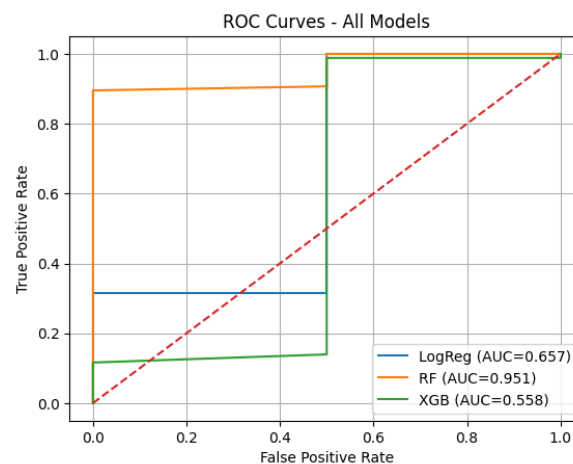


Figure 2. ROC Curves – All Models

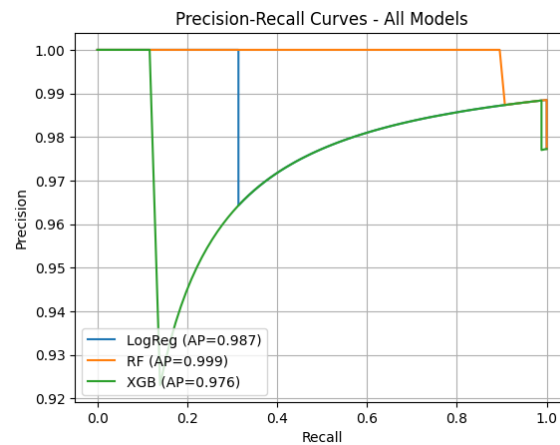


Figure 3. Precision Recall Curves – All Models

Figure 3 shows that Random Forest achieves the highest AUC-PR among the predictive models. Because the positive class represents non-compliant devices, which constitute the majority of the dataset, this result primarily reflects discrimination of the majority class. Therefore, the AUC-PR result is interpreted together with the class-specific metrics in Table 5 when assessing the overall model performance.

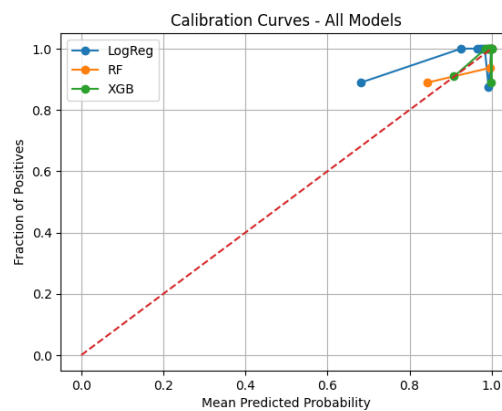


Figure 4. Calibration Curves – All Models

As illustrated in Figure 4, Random Forest appears to follow the diagonal reference line more closely than the other models. However, this pattern should be interpreted cautiously because the test set contains only two compliant cases, making the calibration estimates unstable.

Table 3. Evaluation Recap of Predictive Models and Reference Baselines (Sorted by AUC-PR)

Models	AUC-ROC	AUC-PR	F1@0.5	Prec@0.5	Rec@0.5	Rec@5%	Rec@10%	Rec@20%	Brier
RF	0.950581	0.998	0.994220	0.988506	1.000000	0.046512	0.093023	0.197674	0.013568

Logreg	0.656977	0.986	0.988372	0.988372	0.988372	0.046512	0.093023	0.197674	0.019431
XGB	0.558140	0.976	0.982659	0.977011	0.988372	0.046512	0.093023	0.186047	0.025876
Majority	0.500000	0.977	0.988506	0.977273	1.000000	0.046512	0.093023	0.197674	0.022727
Original KPI Rule	1.000000	1.000	1.000000	1.000000	1.000000	0.046512	0.093023	0.197674	0.000000

Table 4. Class-Specific Sensitivity and Specificity

Model	Compliant Sensitivity	Compliant Specificity	Non-compliant Sensitivity	Non-compliant Specificity
Random Forest	0.500	1.000	1.000	0.500
Logistic Regression	0.500	0.988	0.988	0.500
XGBoost	0.000	0.988	0.988	0.000
Original KPI Rule	1.000	1.000	1.000	1.000

Table 5. Class-Wise Precision, Recall, and F1-Score for Compliant and Non-Compliant Classes (Threshold = 0.5)

Model	Compliant Precision	Compliant Recall	Compliant F1-score	Non-compliant Precision	Non-compliant Recall	Non-compliant F1-score
Random Forest	1.000	0.500	0.667	0.989	1.000	0.994
Logistic Regression	0.500	0.500	0.500	0.988	0.988	0.988
XGBoost	0.000	0.000	0.000	0.977	0.988	0.983
Majority-Class Baseline	0.000	0.000	0.000	0.977	1.000	0.989
Original KPI Rule	1.000	1.000	1.000	1.000	1.000	1.000

Table 6. Repeated Stratified 5-Fold Cross-Validation Results

Model	AUC-ROC (Mean $\pm$ 95% CI)	AUC-PR (Mean $\pm$ 95% CI)	F1-score (Mean $\pm$ 95% CI)
Random Forest	0.8959 $\pm$ 0.0575	0.9989 $\pm$ 0.0006	0.9945 $\pm$ 0.0015
Logistic Regression	0.9467 $\pm$ 0.0285	0.9964 $\pm$ 0.0020	0.9948 $\pm$ 0.0013
XGBoost	0.7338 $\pm$ 0.0857	0.9902 $\pm$ 0.0031	0.9899 $\pm$ 0.0010

Table 3 and 4 show that the Original KPI Rule, used as a rule-based sanity-check baseline, perfectly reproduced the rule-derived labels, whereas the majority baseline failed to identify compliant devices. Among the evaluated predictive models, Random Forest showed the strongest comparative performance across the primary evaluation metrics and achieved the lowest Brier score. However, class-specific performance remained limited because of the very small number of compliant instances. Table 4 reports class-specific sensitivity and specificity, showing that Random Forest and Logistic Regression identified one compliant device, whereas XGBoost identified none.

Table 5 presents class-wise precision, recall, and F1-score for the compliant and non-compliant classes at the default decision threshold of 0.5, offering a more granular view of class-specific performance than the aggregate metrics reported in Table 3. For the majority non-compliant class, all models achieved consistently high scores ($F1 \geq 0.983$), reflecting the extreme class imbalance in the test set. For the minority compliant class, performance diverged sharply across models. Random Forest achieved perfect precision (1.000) but a recall of only 0.500, correctly identifying one of the two compliant devices without any false positives, yielding an F1-score of 0.667. Logistic Regression achieved balanced but lower precision and recall (0.500 each, $F1 = 0.500$), likewise identifying one compliant device but with one false positive. XGBoost and the majority-class baseline both failed to correctly classify any compliant device, resulting in precision, recall, and F1-score of 0.000 for this class. The Original KPI Rule achieved perfect scores across all metrics by construction, since the evaluated labels are themselves derived from this rule and therefore do not constitute an independent validation of predictive performance. These results reinforce that, despite strong aggregate performance, the practical ability of the predictive models to detect compliant devices remains limited and highly sensitive to the small number of compliant instances in the test set.

Table 6 presents the repeated stratified 5-fold cross-validation results over 10 repetitions. Logistic Regression achieved the highest mean AUC-ROC (0.9467) and AUC-PR (0.9989), while Random Forest obtained the highest mean F1-score (0.9948). XGBoost showed comparatively lower mean performance across all three metrics. The variation reflected by the 95% confidence intervals indicates uncertainty across the repeated validation folds, particularly for AUC-ROC. These results complement the single hold-out evaluation and indicate that the comparative advantage of Random Forest is not uniform across all validation metrics.

Figure 5 illustrates the sensitivity of the F1-score to threshold selection, while Figure 6 shows largely overlapping cumulative gain curves, indicating only limited differences in ranking performance among the evaluated models. Therefore, although Random Forest achieved the strongest results on the single hold-out test set, its performance should be interpreted cautiously because the repeated cross-validation results showed different relative performance across evaluation metrics.

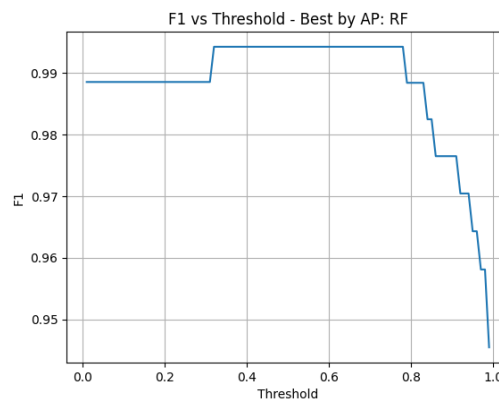


Figure 5. F1 vs Threshold – Best by AP: RF

A lower threshold increases the detection of potential non-compliant devices but may increase the number of devices requiring follow-up, while a higher threshold reduces follow-up workload but may increase missed non-compliance. Thus, threshold selection represents an operational trade-off between detecting non-compliance and limiting follow-up workload. Given the class imbalance and KPI-derived labels, these results should be interpreted as potential operational trade-offs rather than a validated deployment threshold. Figure 6 indicates that all models exhibit nearly identical performance in ranking high-risk devices, with no meaningful separation between the curves.

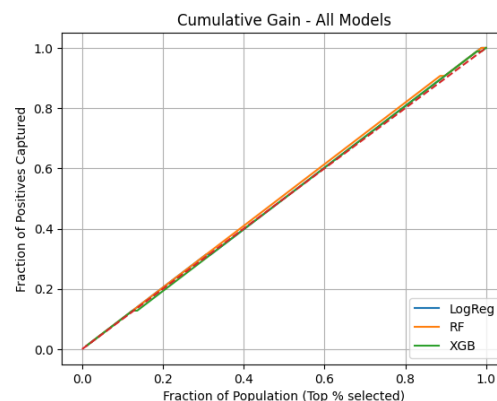


Figure 6. Cumulative Gain – All Models

As shown in Figure 7, all models achieve higher net benefit than the baseline strategies, indicating potential practical utility [32]. Among them, Random Forest generally provides the highest net benefit across a broad range of threshold values. However, given the limited sample size, extreme class imbalance, KPI-derived labeling scheme, and absence of independent external validation, this finding should be interpreted as preliminary

evidence supporting further operational evaluation rather than immediate real-world deployment.

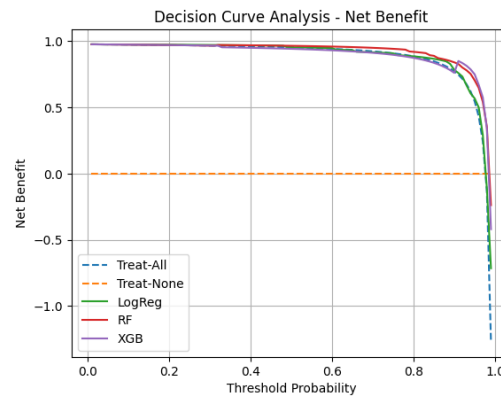
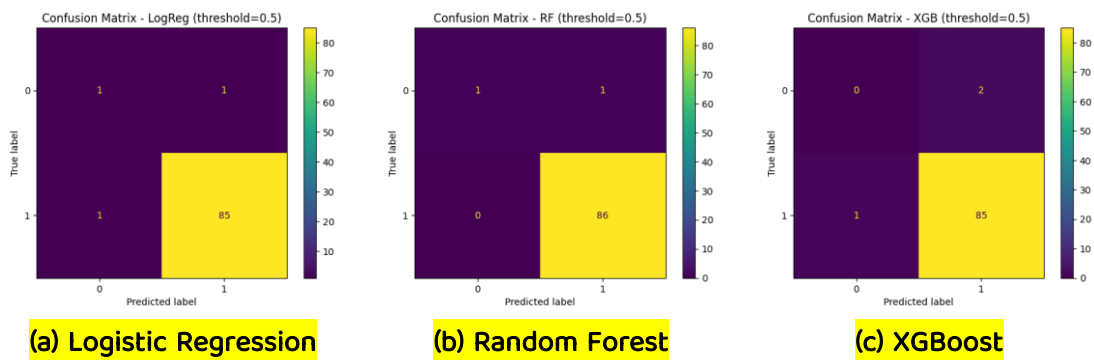


Figure 7. Decision Curve Analysis – Net Benefit

To further support the class-specific results reported in Table 3 and Table 4, Figure 8 presents the confusion matrices at the default decision threshold (0.5) for Logistic Regression, Random Forest, and XGBoost, together with the majority-class baseline and the Original KPI Rule sanity-check baseline. The majority-class baseline classifies every device as non-compliant; it correctly identifies all 86 non-compliant devices but misclassifies both compliant devices, consistent with its Precision@0.5 of 0.977 and Recall@0.5 of 1.000 in Table 3. The Original KPI Rule baseline reproduces the labels perfectly by construction, since the label itself is derived from this rule; its perfect confusion matrix therefore reflects label consistency rather than genuine predictive capability and should not be interpreted as model superiority.



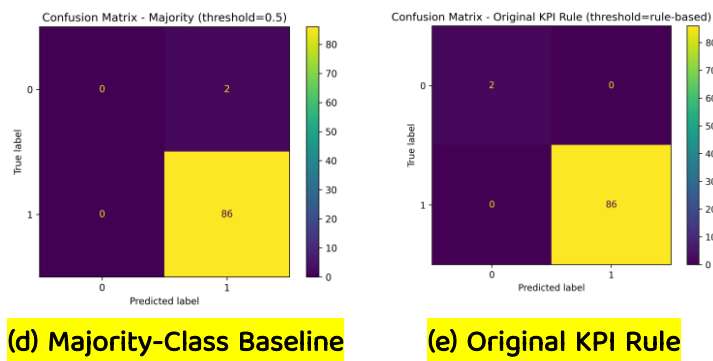


Figure 8. Confusion Matrices – Logistic Regression, Random Forest, XGBoost, Majority-Class Baseline, and Original KPI Rule (threshold = 0.5)

Figure 9 presents the SHAP-based Feature importance for the Random Forest model, computed on the held-out test set, to identify which predictors most influence the model's risk score. The two most influential features are `is_in_acronis` and `days_since_last_backup`, a ranking also confirmed by the impurity-based (Gini) feature importance computed from the same model. Because these two variables are also the exact conditions used to construct the KPI-based compliance label, their dominant contribution should be interpreted cautiously as a symptom of label leakage rather than as evidence of an independently generalizable predictive relationship, a concern examined further in the Discussion section.

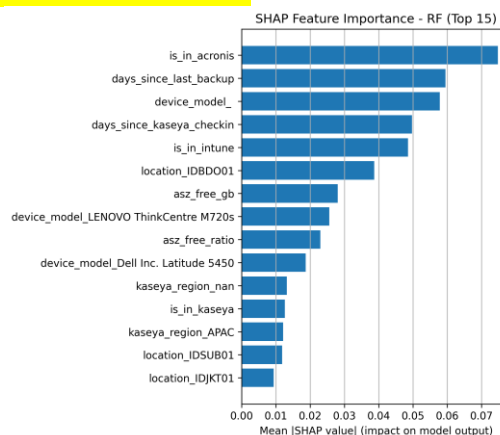


Figure 9. SHAP Feature Importance – Random Forest (Top 15)

4.2. Discussion

The experimental results demonstrate that Random Forest achieves the strongest comparative performance on the single hold-out test set, obtaining the highest AUC-PR, AUC-ROC, F1-score among the evaluated predictive models, together with the lowest

Brier score. However, conclusions regarding probability calibration remain tentative because the test set contains only two compliant instances [33]. The repeated stratified 5-fold cross-validation results provide a more nuanced comparison, with Logistic Regression achieving the highest mean AUC-ROC (0.9467) and AUC-PR (0.9989), while Random Forest achieved the highest mean F1-score (0.9948). Therefore, the strong performance of Random Forest on the hold-out test set should be interpreted as setting-specific rather than as uniform superiority across all validation metrics. Although ensemble-based approaches may be effective in capturing complex patterns of KPI-derived features [4], [27], the cross-validation results show that Logistic Regression can also achieve highly competitive discrimination under repeated validation.

A key issue in interpreting the reported performance is the potential for label leakage and rule replication [34]. The target variable is generated directly from the KPI-based labeling rule, while several predictor variables, including `is_in_acronis` and `days_since_last_backup`, are also used in or are closely related to the conditions that determine the target label. Because the predictors and target are derived from the same operational snapshot, the machine learning models may learn information that closely reflects the deterministic labeling rule. Consequently, the high predictive performance cannot be interpreted as evidence of independently validated real-world compliance prediction [35]. This concern is further corroborated by the SHAP-based feature importance analysis (Figure 9), in which `is_in_acronis` and `days_since_last_backup` emerge as the two most influential predictors, directly mirroring the variables used to construct the label.

The superior performance of Random Forest on the hold-out test set can be attributed to its ability to model non-linear relationships and feature interactions. The KPI-aware features used in this study, such as backup recency, system integration status, and storage-related attributes, inherently exhibit non-linear dependencies. Unlike Logistic Regression, which assumes a linear decision boundary, Random Forest leverages multiple decision trees to partition the feature space more effectively, resulting in improved discrimination between compliant and non-compliant devices [36]. Logistic Regression, despite its simplicity, achieves stable and competitive performance, suggesting that some patterns in the data are linearly separable. Although Logistic Regression achieved lower AUC-ROC on the single hold-out test set, its highest mean AUC-ROC and AUC-PR in

repeated cross-validation indicate that a substantial portion of the discrimination task can also be captured by relatively linear relationships among the selected features. XGBoost shows comparatively lower performance in this study. Although extreme class imbalance may have contributed to this result, the effect cannot be considered conclusive because threshold selection, class weighting, model configuration, and cross-validation performance may also influence the outcome. The repeated cross-validation results show that XGBoost remained comparatively lower than Logistic Regression and Random Forest across the evaluated metrics, although the differences should be interpreted cautiously given the limited number of compliant instances. This highlights the sensitivity of boosting-based methods to severely skewed data distributions [30], [37], [38].

The impact of class imbalance is evident across all models. With only a small number of compliant instances in the test set, the evaluation of minority class performance becomes inherently unstable. While high recall is achieved for non-compliant devices, the models exhibit limited sensitivity in identifying compliant instances, indicating a need for careful interpretation of the results [3], [37]. Further operational validation should compare model predictions with actual backup incidents, audit findings, failed recovery events, and administrator assessments [39].

Operational costs should also be considered when interpreting false positives and false negatives. False positives may lead to unnecessary administrator investigation, remediation, and monitoring workload, whereas false negatives may leave backup issues undetected and increase operational risk. In this context, false negatives may have more serious consequences, although the actual cost depends on organizational procedures and incident impact. Since this study does not quantify these costs, the reported metrics should not be interpreted as evidence that the selected threshold is operationally optimal. Future work should consider administrator workload and the relative costs of missed and unnecessary interventions when determining an appropriate threshold.

The findings also support the main contributions of this study. First, the KPI-based labeling strategy effectively transforms raw operational data into meaningful supervisory signals without requiring manual annotation [22], [23]. Second, the preprocessing and feature engineering pipeline provides a systematic approach for

handling heterogeneous enterprise IT data [13], [21]. Third, the results provide empirical evidence that machine learning models can reproduce KPI-derived proxy labels in the present experimental setting. However, the repeated cross-validation results show that model performance varies across evaluation metrics, with Random Forest performing strongest on the hold-out test set while Logistic Regression achieves the highest mean AUC-ROC and AUC-PR across repeated validation. Therefore, the results do not establish that ensemble-based models are universally more suitable for KPI-driven prediction tasks [9], [11].

From a probabilistic perspective, Random Forest achieves the lowest Brier score, although firm conclusions about its calibration characteristics cannot yet be drawn given the limited number of compliant instances [24], [40]. This is particularly important in enterprise environments, where predicted probabilities may be used to support risk prioritization. However, given the limited number of compliant instances and the absence of external validation, the probability estimates obtained in the present study should be regarded as preliminary and should not yet be interpreted as reliably calibrated for operational decision-making [40]. The model may support risk-based prioritization by combining multiple KPI-derived features into a continuous risk score; however, such operational utility was not directly measured in this study. Furthermore, ranking-based evaluation and decision curve analysis indicate potential operational utility for Random Forest across different threshold settings. However, given the limited separation among the models and the variation observed in repeated cross-validation, the current findings support further operational evaluation rather than immediate real-world deployment [6], [41].

Because the dataset contains device, user-related, backup, platform, and configuration information, privacy and security should be considered in practical deployment, including appropriate access control and secure data handling. Maintainability is also important because changes in KPI thresholds, backup tools, agent versions, or organizational policies may alter feature characteristics, label definitions, and model performance. Such changes can introduce concept drift and reduce the validity of previously trained models. Therefore, the workflow requires version-controlled KPI and labeling rules, monitoring of data and prediction distributions, and periodic recalibration or retraining when significant operational changes occur. Before operational deployment, administrator or domain-

expert validation and independent compliance-audit validation should be conducted to determine whether model outputs and risk rankings correspond to actual backup compliance conditions. These maintenance and validation requirements were not experimentally evaluated in this study and should therefore be addressed in future operational validation.

However, this study has several limitations. The extreme class imbalance limits the evaluation of minority class performance and may affect model generalization. In addition, the dataset is derived from a specific operational environment, which may restrict broader applicability. Future work may explore data balancing techniques, anomaly detection approaches, or temporal modeling to improve robustness and scalability [10], [12], [13]. Model performance may also be affected by concept drift as backup status, device enrollment, agent versions, and operational policies change over time. Periodic monitoring, recalibration, and retraining are therefore required.

4. CONCLUSION

This study addresses the limitations of traditional rule-based monitoring systems in enterprise IT environments, which are often static, reactive, and unable to handle the complexity and scale of modern infrastructures. To overcome these challenges, a KPI-aware machine learning approach is proposed for predicting backup non-compliance. The main contributions are threefold: (1) a KPI-based labeling mechanism that enables automatic label generation without manual annotation, (2) a robust preprocessing and feature engineering pipeline for heterogeneous enterprise IT data, and (3) a comprehensive evaluation of machine learning models under highly imbalanced conditions. Together, these contributions provide a preliminary workflow for KPI-based backup compliance monitoring and risk prioritization.

Compared with Logistic Regression and XGBoost, Random Forest demonstrates the strongest comparative classification performance and obtains the lowest Brier score within the evaluated test set. Nevertheless, no firm conclusion regarding probability calibration can be drawn because the test set contains only two compliant instances. These findings suggest the preliminary feasibility of the proposed workflow under the current KPI-derived labeling scheme. More specifically, the results answer the research

questions by showing that machine learning models can reproduce KPI-derived backup compliance labels, with Random Forest achieving the strongest discrimination performance among the evaluated models under extreme class imbalance. However, the cumulative gain analysis indicates only limited differences in ranking performance among the models, and therefore no clear ranking superiority can be established. The probabilistic outputs produced by the Random Forest model may nevertheless provide potential operational value by supporting risk-based prioritization of devices rather than relying solely on binary rule-based decisions. However, this potential operational benefit has not yet been validated against independent compliance audit labels, and the reported performance may be overestimated because the target labels are derived from KPI features that are also available to the learning process.

This study is subject to several limitations, including the relatively small sample size, the use of data collected from a single organizational environment, severe class imbalance (the test set contained only 2 compliant devices out of 88 instances, limiting the stability of class-wise evaluation), and the absence of temporal validation. Because the labels and predictors are derived from the same operational snapshot, the current workflow reproduces the current rule-derived compliance status rather than predicting non-compliance before a policy violation occurs. Consequently, the reported results should be interpreted within the scope of the current experimental setting, and leakage-controlled experiments remain necessary to confirm whether the machine learning models provide predictive value beyond the deterministic KPI rule itself.

Future work should therefore prioritize independent compliance audit labels, larger multi-organization datasets to obtain a more stable and generalizable estimate of class-wise performance, temporal prediction capability, leakage-controlled feature sets, administrator validation, and real-time operational deployment testing.

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